    - The agent explores actions randomly and updates the Q-Table based on the observed rewards and future Q-values.
    - Over multiple episodes, the Q-Table converges to an optimal policy.
5. **Policy Extraction**:
    - Once training is complete, the agent can determine the best action for each state by selecting the action with the highest Q-value.

Applications of Q-Learning

1. **Pathfinding and Navigation**:
    - Robots or autonomous agents can learn to navigate mazes or avoid obstacles efficiently.
2. **Game AI**:
    - Q-Learning has been applied to simple games like Tic-Tac-Toe and advanced games such as Atari classics.
3. **Resource Optimization**:
    - Allocate resources dynamically in systems like cloud computing or network routing.
4. **Dynamic Pricing**:
    - Optimize pricing strategies in e-commerce or retail environments.

Advantages of Q-Learning

- **Model-Free**: Does not require knowledge of the environment's dynamics.
- **Generalizable**: Can be applied to various domains with discrete states and actions.
- **Asynchronous Updates**: Learns effectively without needing all states and actions to be visited in every iteration.